Week 8 — AI Spec Development & Capstone

> *"Eight weeks of muscle. Five days of proof."*

Week 8 is the capstone. Every triad works the same **capstone subject** — the **Holocron** enterprise string-management problem, supplied as a problem brief at the end of Week 7 — and produces a compressed-but-complete artifact set, anchored on the new technique introduced this week: **AI Spec Development** — using AI to draft the integrated technical spec that ties PRD / TCD / TMD outputs together, with full validation discipline.

Friday is **final artifact presentations**. The cohort + instructors score on a rubric that touches every muscle from Weeks 1–7.

What's different about Week 8

- **The subject is fixed: Holocron.** Every triad gets the same problem brief and writes its own requirements from it — nothing is pre-specified.
- **Compression.** The 5-artifact set produced over Weeks 3–7 gets re-produced in **one week** for the new subject. This tests whether the muscles transfer.
- **AI Spec Development is new.** Day 1 introduces the technique; Day 3 applies it.
- **Friday is presentations.** Each triad has 15 minutes + 5 min Q&A.
- **The tone is summative, not introductory.** This week shows whether the academy worked.

Learning outcomes

By Friday afternoon, each participant can:

1. Apply the **AI Spec Development pattern** — a structured prompt sequence that produces an engineering-ready integrated technical spec, with explicit validation against the PRD/TCD/TMD inputs.
2. Re-produce the artifact set (PRD-light / TCD-light / TMD-light / SEP-light / DP-light) for the capstone subject in **5 days**, applying the muscles built across Weeks 1–7 end-to-end.
3. Run **technical logic validation** — checking metric coherence, architectural consistency, data-API alignment, and trade-off honesty across the artifact set before shipping.

4. Present the integrated artifact set in a **15-minute readout** that lands with both engineering and business audiences.
5. Reflect on the **8-week arc** — what stuck, what to deepen, what habits to keep.

Daily map

Day	Topic	Key artifact produced
1	AI Spec Development	AI Spec template + Holocron scope slice locked
2	Capstone discovery + compressed PRD	Capstone PRD-light
3	Capstone architecture + AI Spec drafted	TCD-light + TMD-light + AI Spec v1
4	Technical logic validation + finalization	Validated, integrated capstone artifact set
5	Final artifact presentations + course closure	Presented capstone; course complete

Daily cadence (Mon–Thu — Friday differs)

Clock	Block	Mix
09:00 – 09:15	Opening & objectives	Instructor
09:15 – 10:30	Teaching Block 1 + Activity 1	~25 min teach / 50 min triad work
10:30 – 10:45	**Break**	
10:45 – 12:00	Teaching Block 2 + Activity 2	~25 min teach / 50 min triad work
12:00 – 13:00	**Lunch**	
13:00 – 14:15	Teaching Block 3 + Activity 3	~25 min teach / 50 min triad work
14:15 – 14:30	**Break**	
14:30 – 16:00	Teaching Block 4 + Activity 4 + Wrap	~25 min teach / 45 min triad / 20 min wrap

Friday cadence (presentations)

For 6 triads (typical cohort), Friday is structured as:

Clock	Block	What happens
09:00 – 09:30	Opening; presentation order; rubric; final tweaks	Instructor + triads
09:30 – 10:50	**Triad 1 + Triad 2 presentations** (15 + 5 min Q&A each, plus 10 min flex)	Cohort scores
10:50 – 11:05	Break	
11:05 – 12:25	**Triad 3 + Triad 4 presentations**	
12:25 – 13:25	Lunch	
13:25 – 14:45	**Triad 5 + Triad 6 presentations**	
14:45 – 15:15	Cohort retrospective	What worked; what to deepen
15:15 – 16:00	Course closure: post-assessments, certificates, wrap	Instructor-led

If the cohort is smaller, Friday compresses; if larger, the morning extends and break shifts.

The capstone artifact set (compressed)

Each triad ships a **compressed but complete** version of the 5-week artifact set:

```
capstone/
├── PRD-light.md          (sections 1–5 + 6 AC + 7 NFR — shorter than the 11-section Week-3 version)
├── TCD-light.md          (stance + integration + threat-model summary + SLOs + top trade-offs)
├── TMD-light.md          (data + cloud + API + sequence + baselines)
├── SEP-light.md          (stakeholder map + 1 trade-off brief + simulated negotiation outcome)
├── DP-light.md           (outcome map + backlog skeleton + tracking + 1 experiment)
└── AI-Spec.md            (NEW THIS WEEK — integrated engineering-ready spec)
```

Compression is the constraint. Every artifact must fit on **2 pages** or less. The discipline is **what to cut**, not what to add.

Triads

Same triads from Weeks 1–7. The PRD authors → TCD authors → TMD authors → SEP authors → DP authors → AI-Spec authors → presenters.

Friday presentation rubric

Each triad's 15-min presentation is scored on:

Dimension	Weight	Exemplary
Customer + problem clarity	15%	Engineer or stakeholder could understand the customer's pain in 90 seconds
Integrated artifact set	15%	All 6 artifacts referenced; references are consistent
AI Spec quality	20%	Engineering-ready; integrates PRD/TCD/TMD outputs; provenance log included
Trade-off honesty	15%	At least 2 trade-offs named with cost + revisit trigger
Outcome thinking	10%	NS / Tier-Sheet / leading indicator vocabulary used naturally
Delivery readiness	10%	DP-light shows backlog, tracking, one bottleneck experiment
Presentation craft	15%	All triad voices speak; time-managed; questions answered with curiosity not defensiveness

Bridge — what closes when the academy closes

- Post-assessments (Product Management + Data Literacy) administered after presentations
- Each participant declares **two muscles to keep deepening** in their next role
- Each participant declares **one habit to retire**
- Cohort retrospective captures patterns for the next academy cohort